

Full Stack Development with MERN

Project Documentation format

1. Introduction

Project Title: Automated Network Request Management

2. Project Overview

Purpose:

- Automate the network request management process.
- Reduce manual effort and approval time.
- Improve request tracking and transparency.
- Ensure timely assignment and resolution of network issues.

Features:

- Network Request Form
- Automated Approval Workflow
- Auto Assignment to Network Team
- Status Tracking
- Email Notifications
- Reports and Dashboards

3. Architecture

- **Frontend:** ServiceNow Forms, Service Catalog and List Views.
- **Backend:** Flow Designer, Business Rules, Script Includes, and Notifications.
- **Database:** Custom ServiceNow table named "Network Data" to store request information. The table contains fields such as Request Number, Employee Name, Department, Network Issue Type, Priority, Status, Assigned To, and Resolution Date. ServiceNow manages all database operations, including record creation, retrieval, updates, and reporting.

4. Setup Instructions

- **Prerequisites:** ServiceNow Personal Developer Instance (PDI), System Administrator Role, Internet Browser
- **Installation:**
 - i) Create the Network Data table.
 - ii) Add required fields.
 - iii) Design the request form.
 - iv) Configure Flow Designer.
 - v) Create Business Rules and Notifications.
 - vi) Test the application.

5. Folder Structure

Client:

The frontend consists of:

- **Service Catalog** – User interface for submitting network requests.
- **Forms** – Displays and edits request details.
- **List Views** – Shows all network requests.
- **Dashboards** – Displays reports and request statistics.

Server:

The backend consists of:

- **Custom Table (Network Data)** – Stores all network request records.
- **Flow Designer** – Automates approval and assignment workflows.
- **Business Rules** – Executes server-side automation.
- **Script Includes** – Contains reusable server-side logic.
- **Notifications** – Sends email alerts for request updates.
- **Access Control (ACLs)** – Manages user permissions and security.

6. Running the Application

- Log in to the ServiceNow instance using valid credentials.
- Navigate to the Service Catalog.
- Select the Network Request catalog item.
- Fill in the required details and submit the request.
- The automated workflow processes the request through approval, assignment, and resolution.

7. API Documentation

Operation	Method	Parameters	Example Response
Create Network Request	POST	Employee Name, Department, Network Issue Type, Description, Priority	Request created successfully with Request Number.
View network requests	GET	Request Number	Returns the list of submitted network requests
Update Request status	PUT	Request Number, Status, Assigned to	Request status updated successfully
Delete Request (only access to admin)	DELETE	Request Number	Request deleted successfully

8. Authentication

- **Authentication:** Users log in using their ServiceNow credentials through the platform's secure login system.
- **Authorization:** Access is controlled using Role-Based Access Control (RBAC). Different roles (such as Admin, ITIL User, and Employee) determine the features and data each user can access.
- **Access Control:** **Access Control Lists (ACLs)** are used to restrict access to tables, records, and fields based on user roles.

- **Sessions:** ServiceNow manages user sessions automatically, allowing secure access until the user logs out or the session expires.
- **Tokens:** No custom authentication tokens are implemented, as authentication is managed internally by the ServiceNow platform.

9. User Interface

Include screenshots of:

- Service Catalog

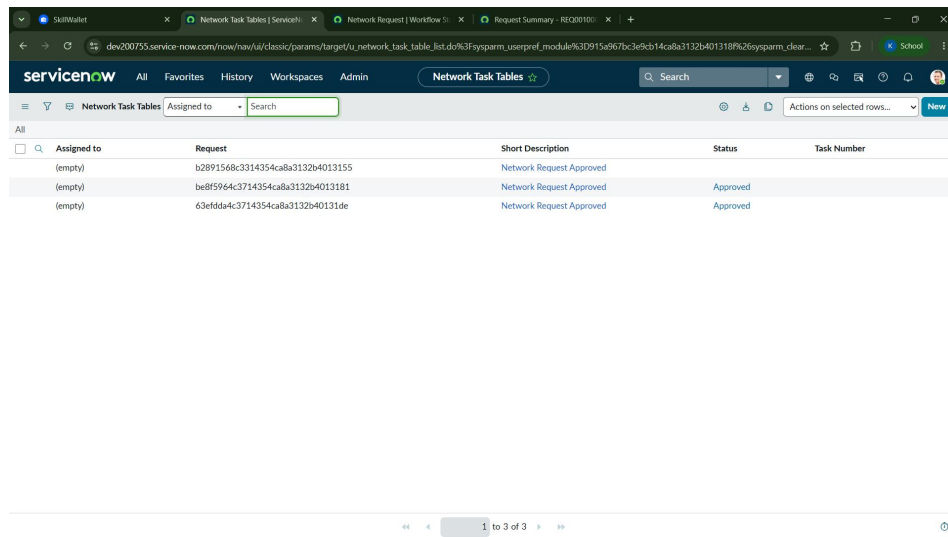
The screenshot shows the ServiceNow 'Network Request' form. At the top, there is a red banner for 'Employee Center'. Below it, the navigation bar includes 'Home', 'Service Catalog', 'Standard Changes', 'Network Standard Changes', and 'Network Request'. The form itself has a header 'Network Request' and 'Network Service Management'. It includes fields for 'Opened on behalf of' (Aileen Mottern), 'Email ID' (aileen.mottern@example.com), 'User Name' (Aileen Mottern), and 'Phone Number'. There is a 'Proof of Attachment' section with an 'Upload' button. A right-hand sidebar contains a 'Quantity' dropdown (set to 1), 'Delivery Time' (0 Days), and buttons for 'Add to Cart', 'Save as Draft', and 'Order Now'. At the bottom, there are radio buttons for 'New' and 'Relocation'.

- Network Request Form

The screenshot shows a list of emails in the ServiceNow interface. The table has columns for 'Created', 'Recipients', 'Subject', 'Type', 'Notification type', and 'User ID'. The data is filtered by 'Created' and shows 16 rows of email notifications related to network requests.

Created	Recipients	Subject	Type	Notification type	User ID
2026-06-27 02:02:19	admin@example.com	[PA VSC] Daily Data Collection completed with error	send-ready	SMTP	(empty)
2026-06-27 01:00:09	aileen.mottern@example.com	Restocking Request For APC 42U 3100 SP2 NetShelter Rack	send-ready	SMTP	(empty)
2026-06-27 01:00:10	aileen.mottern@example.com	Restocking Request For Dell Inc. PowerEdge M710HD Blade Server	send-ready	SMTP	(empty)
2026-06-27 01:53:59	abraham.lincoln@example.com	Network Task Created	send-ready	SMTP	(empty)
2026-06-27 02:18:19	admin@example.com	Request REQ0010009 was approved	send-ready	SMTP	(empty)
2026-06-27 02:16:29	admin@example.com	Request REQ0010008 was created	send-ready	SMTP	(empty)
2026-06-27 01:50:21	abraham.lincoln@example.com	Network Request Submitted Successfully	send-ready	SMTP	(empty)
2026-06-27 02:16:39	admin@example.com	Request REQ0010008 was approved	send-ready	SMTP	(empty)
2026-06-27 00:00:12	admin@example.com	Daily job to fetch Email Indicator Data and Email Notifications created completed with error	send-ready	SMTP	(empty)
2026-06-27 01:50:20	admin@example.com	Request REQ0010007 was approved	send-ready	SMTP	(empty)
2026-06-27 02:18:15	aileen.mottern@example.com	Network Request Submitted Successfully	send-ready	SMTP	(empty)
2026-06-27 01:50:20	admin@example.com	Request REQ0010007 was created	send-ready	SMTP	(empty)
2026-06-27 01:57:07	abraham.lincoln@example.com	Network Request Completed	send-ready	SMTP	(empty)
2026-06-27 02:18:19	admin@example.com	Request REQ0010009 was created	send-ready	SMTP	(empty)
2026-06-27 01:00:09	aileen.mottern@example.com	Restocking Request For Fujitsu 1TB Hybrid Solid State Drive	send-ready	SMTP	(empty)
2026-06-27 02:16:34	abel.tuter@example.com	Network Request Submitted Successfully	send-ready	SMTP	(empty)

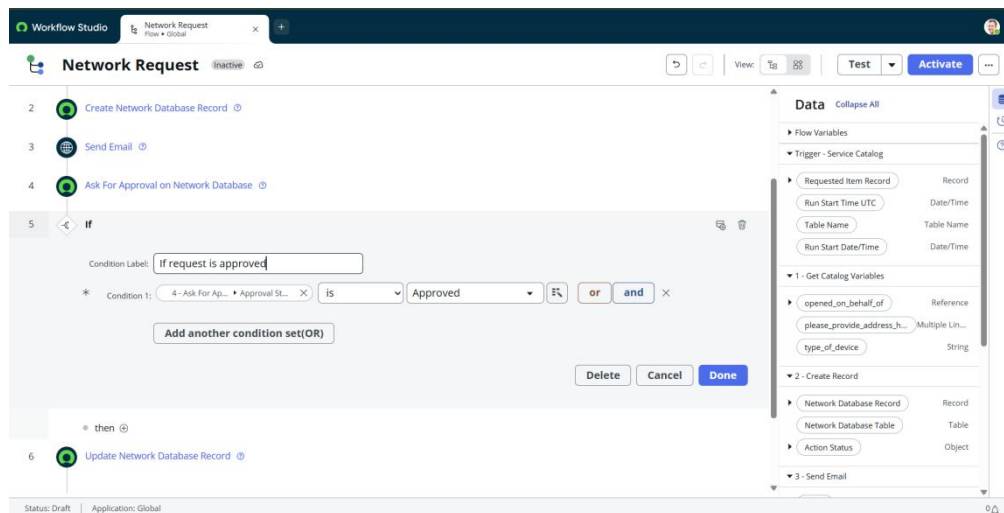
- Network Data Table



Assigned to	Request	Short Description	Status	Task Number
(empty)	b2891568c3314354ca8a3132b4013155	Network Request Approved	Approved	
(empty)	be8f5964c3714354ca8a3132b4013181	Network Request Approved	Approved	
(empty)	63efdda4c3714354ca8a3132b40131de	Network Request Approved	Approved	

10. Testing

- Functional Testing



Network Request Workflow Studio

Step 5: If

Condition Label: If request is approved

Condition 1: 4 - Ask For Ap... Approval St... is Approved

Buttons: Delete, Cancel, Done

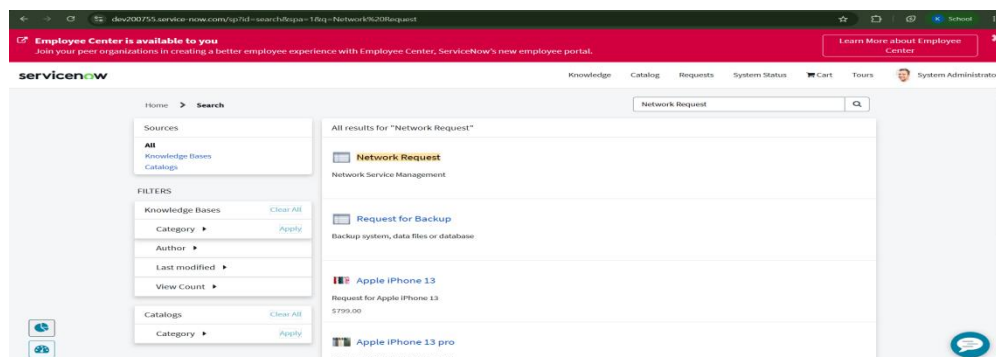
Step 6: Update Network Database Record

Status: Draft | Application: Global

Data Panel:

- Flow Variables
- Trigger - Service Catalog
- Requested Item Record
- Run Start Time UTC
- Table Name
- Run Start DateTime
- 1 - Get Catalog Variables
- opened_on_behalf_of
- please_provide_address_h...
- type_of_device
- 2 - Create Record
- Network Database Record
- Network Database Table
- Action Status
- 3 - Send Email

- Form Validation



Employee Center is available to you

Join your peer organizations in creating a better employee experience with Employee Center, ServiceNow's new employee portal.

servicenow

Home > Search

Search: Network Request

Filters:

- Knowledge Bases
- Category
- Author
- Last modified
- View Count
- Catalogs

All results for "Network Request"

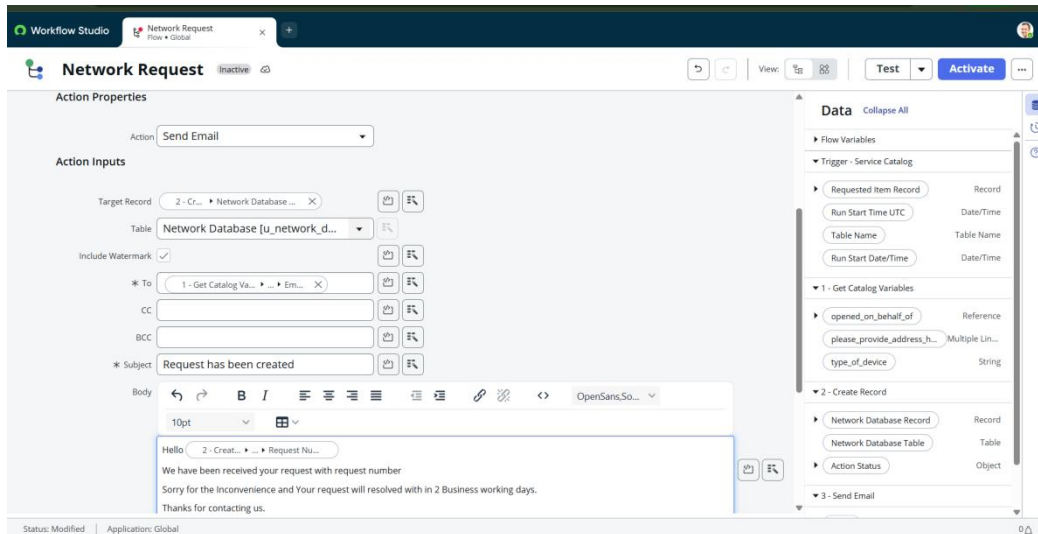
- Network Request**
- Network Service Management
- Request for Backup**
- Backup system, data files or database
- Apple iPhone 13**
- Request for Apple iPhone 13
- \$799.00
- Apple iPhone 13 pro**
- Request for Apple iPhone 13 pro

- Workflow Testing

Workflow Studio interface showing the configuration for a 'Service Catalog' trigger. The trigger is set to 'Service Catalog'. Below the trigger, there is an 'Advanced Options' dropdown menu. At the bottom right, there are buttons for 'Delete', 'Cancel', and 'Done'.

Workflow Studio interface showing the configuration for a 'Create Network Database Record' action. The action is set to 'Create Record'. The 'Action Inputs' section shows a table with columns for 'Request Number', 'Requested For', 'Work Status', 'Assignment Group', 'Date of Enquiry', 'Device Details', and 'Customer Address'. Each column has a dropdown menu for selecting a value. The 'Data' panel on the right shows the data structure for the 'Create Record' action, including 'Requested Item Record', 'Run Start Time UTC', 'Table Name', 'Run Start Date/Time', 'opened_on_behalf_of', 'please_provide_address_here', 'type_of_device', 'Network Database Record', 'Network Database Table', and 'Action Status'.

Workflow Studio interface showing the configuration for a 'Get Catalog Variables' action. The action is set to 'Get Catalog Variables'. The 'Action Inputs' section shows a dropdown menu for selecting a value. The 'Data' panel on the right shows the data structure for the 'Get Catalog Variables' action, including 'Requested Item Record', 'Run Start Time UTC', 'Table Name', 'Run Start Date/Time', 'opened_on_behalf_of', 'please_provide_address_here', 'type_of_device', 'Network Database Record', 'Network Database Table', and 'Action Status'.



• Notification Testing

Flow executions ¹

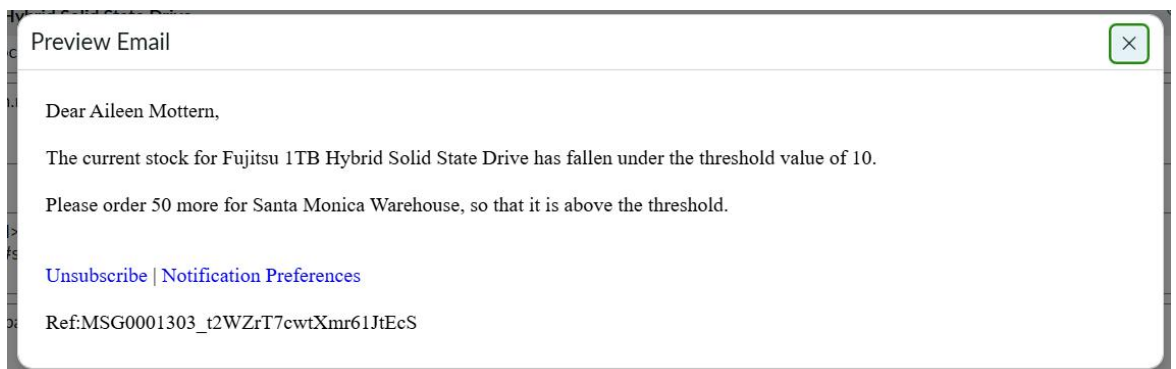
Last refreshed about an hour ago.

	Name ▲	State	Runtime	Created by	Created	Sys ID
☐	Network Request	Complete	1,793	system	2026-06-26 23:53:57	78eebce086d

Showing 1-1 of 1

1

10 rows per page



• User Acceptance Testing (UAT)

servicenow

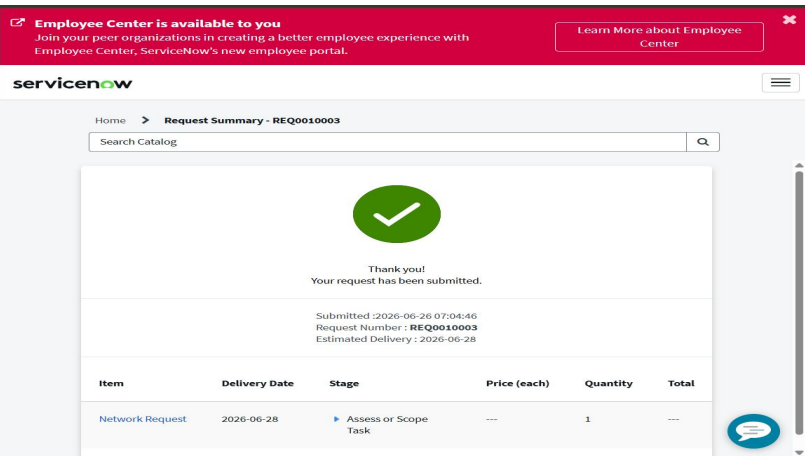
Network Task Tables

Assigned to	Request	Short Description	Status	Task Number
(empty)	k2891568c3314354ca3132b4013155	Network Request Approved	Approved	
(empty)	be8f5964c3714354ca3132b4013181	Network Request Approved	Approved	
(empty)	63ef6da4c3714354ca3132b40131de	Network Request Approved	Approved	

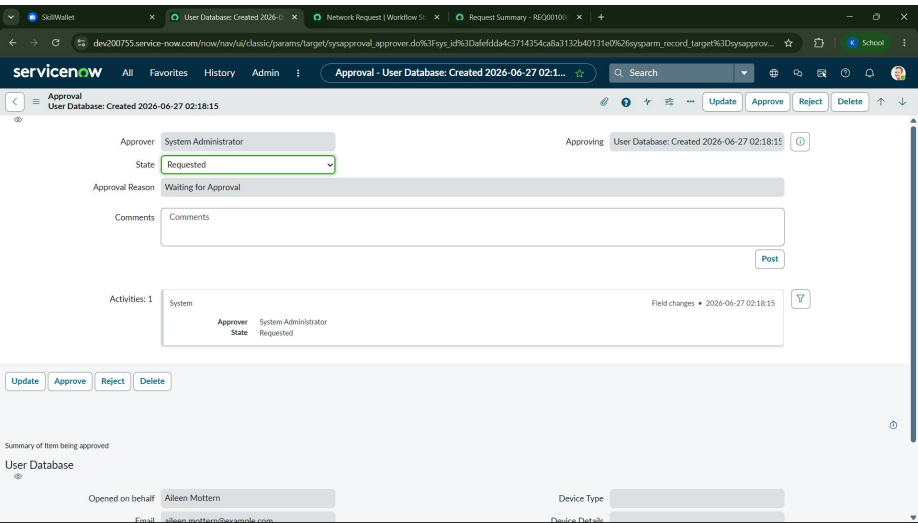
11. Screenshots or Demo

Include:

- Request Form



- Approval Workflow



- Network Data Table

State	Approver	Comments	Approval for	Created
Approved	System Administrator		(empty)	2026-06-27 01:50:21
Requested	System Administrator		(empty)	2026-06-27 02:18:15
Approved	System Administrator		(empty)	2026-06-27 01:53:59
Requested	System Administrator		(empty)	2026-06-27 02:16:34

- Completed Request

Assigned to	Request	Short Description	Status	Task Number
(empty)	12891568c3314354ca8a3132b4013155	Network Request Approved		
(empty)	be8f9f64c3714354ca8a3132b4013181	Network Request Approved	Approved	
(empty)	63efdd84c3714354ca8a3132b40131de	Network Request Approved	Approved	

12. Known Issues : SMTP configuration required for email notifications , External integrations require additional setup.

13. Future Enhancements

- AI-Based Request Categorization:** Automatically classify and prioritize network requests using AI to reduce manual effort and improve response time.
- Virtual Agent Integration:** Enable users to submit requests, check request status, and receive instant assistance through the ServiceNow Virtual Agent.
- Real-Time Analytics Dashboards:** Provide live dashboards to monitor request volume, SLA compliance, resolution time, and overall network support performance.